

Assessing the Impact of Visual and Auditory Perspectives on Performance and Pseudo-Haptics in a Virtual Drilling Task

Sandy Abdo*
Ontario Tech University,
Oshawa, Canada

Bill Kapralos†
Ontario Tech University,
Oshawa, Canada

KC Collins‡
Carleton University,
Ottawa, Canada

Adam Dubrowski§
Ontario Tech University,
Oshawa, Canada

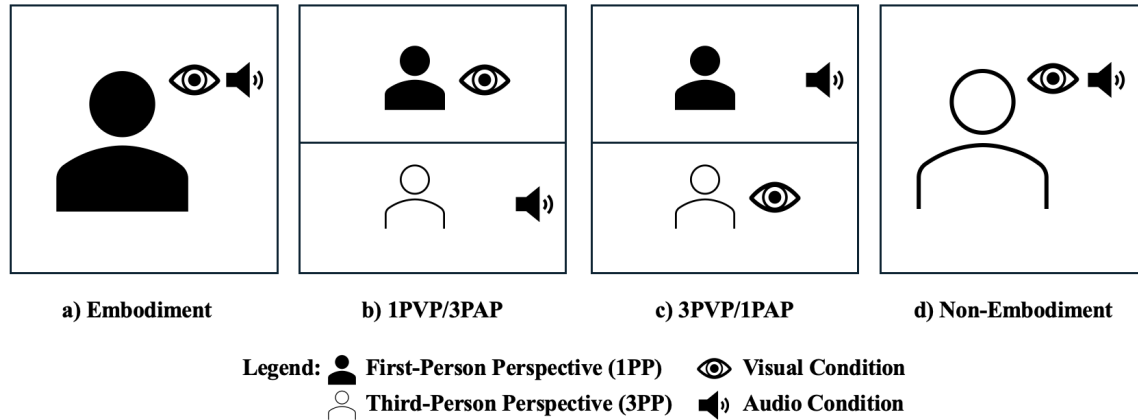


Figure 1: Visual and auditory perspective conditions: **a) Embodiment**, where both visual and auditory conditions are in the first-person perspective; **b) 1PVP/3PAP**, where the visual condition is in the first-person perspective (1PVP) and the auditory condition is in the third-person perspective (3PAP); **c) 3PVP/1PAP**, where the visual condition is in the third-person perspective (3PVP) and the auditory condition is in the first-person perspective (1PAP); and **d) Non-Embodiment**, where both visual and auditory conditions are in the third-person perspective.

ABSTRACT

Psychomotor skill development has been neglected in immersive virtual learning environments (iVLEs) due to the difficulties in simulating the sense of touch (haptics). Although pseudo-haptics that uses multi-modal interaction to simulate the sense of touch without relying on costly and inaccessible haptic devices, has emerged as a promising field, research has often overlooked the importance of embodiment, the sense of being present and controlling a body, with respect to pseudo-haptics. In this work-in-progress paper, we propose an experiment that examines the interplay between visual and auditory perspectives and their effect on haptic perception via pseudo-haptics and on psychomotor performance in a virtual reality drilling task. We aim to bridge the gap in in virtual-based psychomotor skill training research by providing valuable insights into the role of embodiment and pseudo-haptics in enhancing simulation-based training, particularly in scenarios where access to high-fidelity haptic devices is limited.

Index Terms: Virtual reality, auditory feedback, users studies, user centered design, human computer interaction (HCI)

1 INTRODUCTION

Immersive virtual learning environments (iVLEs), particularly those using virtual reality (VR), have the potential to enhance cog-

nitive and affective skills yet face challenges in their application to psychomotor skills development due to limitations in replicating the sense of touch (haptics), which is essential for psychomotor skill development [14, 13, 15]. Accurately simulating touch requires real-time perception of pressure, temperature, movement, texture, and vibration, typically necessitating high-end haptic devices that are costly and therefore often restricted to university laboratories and research centers [15, 1]. While consumer-level haptic devices exist, they generally offer simple feedback, such as vibration from actuators found in video game controllers or mobile phones [1, 15]. More advanced consumer-level devices, such as the Novint Falcon, have been released but lack the fidelity and range of motion needed to simulate tasks realistically [13]. To overcome these limitations associated with haptic devices, particularly their high cost, the field of pseudo-haptics has emerged. Pseudo-haptics uses cross-modal illusions, primarily visual cues, to simulate the perception of haptics, and has been shown to convey properties such as friction, stiffness, weight, force, and texture [6, 20].

In addition to the critical role of touch, psychomotor skill development also depends on embodiment—the sense of owning and controlling a body within a virtual environment. Embodiment is fundamental for psychomotor skill acquisition, as motor skills require the physical presence to initiate, govern, and control body movement [9, 2]. However, this key aspect is often overlooked in psychomotor learning [7].

Kilteni et al. define embodiment as comprising three key sub-components: (i) self-location, (ii) agency, and (iii) ownership [9]. Self-location pertains to the perception of being situated within a body and understanding its spatial positioning within an environment. This sub-component, heavily reliant on vision and the individual's perspective, is distinguished by its self-aware nature, making it a critical aspect of embodiment [9, 19]. Agency, on the other

*e-mail: sandy.abdo@ontariotechu.net

†e-mail: bill.kapralos@ontariotechu.ca

‡e-mail: KCCollins@cunet.carleton.ca

§e-mail: adam.dubrowski@ontariotechu.net

hand, refers to the capacity for voluntary and conscious control over bodily movements. Lastly, ownership—the most extensively studied of the three sub-components—relates to the sensation that an artificial object is perceived as part of one’s own body [9, 19].

In our previous work, we explored the integration of pseudo-haptics with ownership during a virtual drilling task where participants were tasked with drilling through a virtual block of wood from a first-person perspective (in a non-immersive VR environment). Forty participants were randomly assigned to either an embodiment group (presence of a virtual hand) or a non-embodiment group (absence of a virtual hand) as shown in Figure 2. Results indicated that combining pseudo-haptics with ownership enhanced task performance, as evidenced by reduced completion times and fewer numbers of attempts to completing the virtual drilling task.

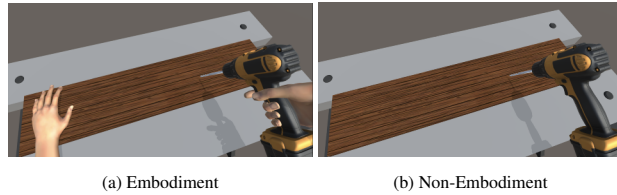


Figure 2: Visual conditions used in the study.

In this paper, we propose a study to investigate how changes in visual and auditory perspectives (first-person vs. third-person) affect perceived haptic sensations and performance during a virtual drilling task in VR. Specifically, we aim to address the following research question: What effect does a change in visual and auditory perspectives have on the perception of haptic feedback, presence and performance in a virtual drilling task in VR?

According to Cheng et al., incorporating auditory perspective into visual perspective, particularly third person, could enhance agency, a sub-component of embodiment [5]. However, unlike its visual counterpart, auditory perspective is less explored [5, 4], and greater work is needed to understand the interplay between visual and auditory perspectives, particularly in how they affect psychomotor performance and haptic perception.

By examining the interplay between perspectives and pseudo-haptics, this work provides valuable insights for developing more precise and user-centered simulations, making a significant contribution to digital twin research. Digital twins rely on accurate, immersive, and interactive virtual representations to replicate their physical counterparts [11]. This research not only enhances training efficacy in fields such as medicine but also expands the potential of digital twins to deliver personalized, adaptive, and accessible solutions for addressing complex real-world challenges.

2 EXPERIMENTAL METHOD

To investigate the potential differences between first-person and third-person perspectives on pseudo-haptics and performance in a VR-based virtual drilling task, a between-subjects experiment will be conducted with an anticipated 40 participants. This approach aims to avoid carryover effects and biases from exposure to multiple conditions. Although a within-subjects design is effective in controlling individual variability, it is not feasible in this context, as repeated exposure may skew physiological responses and obscure the true effects of the intervention. Given the pilot nature of this study, the between-subjects design is essential for maintaining the integrity of the intervention and providing clearer insights into its effects.

The study will take place in-person at a research laboratory at Ontario Tech University in Oshawa, Canada, facilitated by a researcher who will guide participants and ensure the smooth completion of each session.

Before the study begins, 40 participants will be randomly assigned to one of four conditions (10 participants per group) in a 2×2 factorial design, which includes one visual perspective (VP) and one auditory perspective (AP): (i) *Embodiment*, where the drilling task is conducted from a first-person perspective both visually and aurally (Fig.1a), (ii) *1PVP/3PAP*, where the drilling task is conducted visually in the first-person perspective (1PVP) but aurally in the third-person perspective (3PAP) (Fig.1b), (iii) *3PVP/1PAP*, where the drilling task is conducted visually in the third-person perspective (3PVP) but aurally in the first-person perspective (1PAP) (Fig.1c), and (iv) *Non-Embodiment*, where the drilling task is conducted from a third-person perspective both visually and aurally (Fig.1d). Participants will also receive a brief overview of the experiment, including its purpose and task requirements. The virtual drilling task will be conducted in a virtual environment using an Meta Quest 3 in standalone mode, controlled via a VR controller. Pressing the trigger button will activate the drill, while releasing it will stop the drill.

The participants’ objective is to drill to a specific depth of 15 cm in a virtual block of wood. Participants must achieve the correct depth five consecutive times; failure to do so will result in restarting the task from the first attempt. A sneak peek at the visual conditions (1PP vs. 3PP) is shown in Figure 3. To enhance the multi-modal nature of the experience, kinesthetic feedback will be included as participants physically manipulate the VR controller to control the drill.

Upon completing the session, participants will exit the virtual environment and complete a series of questionnaires on the experimenter’s computer. The first two levels of the Kirkpatrick Model will be used to evaluate the intervention: Reaction and Performance [10]. Reaction, including haptic perception, will be evaluated using the Michigan Standard Simulation Experience Scale, NASA Task Load Index (TLX), and the System Usability Scale (SUS) [17, 8, 16]. Presence will be measured using the Igroup Presence Questionnaire (IPQ), a validated questionnaire [3] and psychomotor performance will be assessed through task completion time, number of attempts, and drilling accuracy (actual vs. target depth).

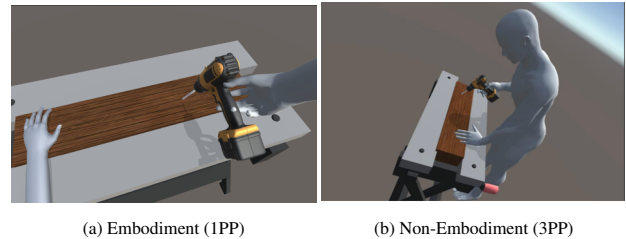


Figure 3: Visual conditions of the study.

3 SIGNIFICANCE

The work introduced in this paper investigates the interplay between perspective (first-person and third-person perspectives) and pseudo-haptics in a VR environment, offering insights into their combined effects on psychomotor skill development. Perspective plays a pivotal role in shaping how users perceive and interact with virtual environments, influencing their sense of embodiment (via self-location), spatial awareness, and task performance [5, 9, 18].

For instance, a first-person perspective aligns the user’s self-location with the virtual avatar, potentially providing a more immersive and realistic experience that enhances precision, reaction times, and potentially the perception of tactile feedback through pseudo-haptics [9, 5, 18]. In contrast, a third-person perspective may offer a broader view of the task environment but could compromise the user’s ability to feel fully present or effectively control

the virtual body [5, 9, 12]. Identifying the optimal perspective is crucial for tailoring iVLEs to specific skills and contexts, ensuring learners achieve the highest levels of accuracy, efficiency, and skill transfer to real-world applications. This knowledge will help design VR tools that are intuitive and engaging, making them more effective and accessible across diverse fields, including medical training.

Moreover, the findings of this work will have significant implications for creating accessible and high-quality iVLEs, particularly in fields such as medicine, where psychomotor skill training is essential. By uncovering how perspective and pseudo-haptics influence task performance and haptic perception, this research not only addresses a gap in the literature but also paves the way for more immersive and effective simulation-based training tools. Additionally, the insights gained from this study contribute to the personalization of digital twin technology, enabling the creation of adaptive iVLEs that can be tailored to individual needs and learning styles. This personalization enhances the training experience, allowing for more precise and effective simulations that closely mirror real-world tasks, ultimately improving skill development and transferability to practical applications.

Overall, this work provides valuable insights for designing more intuitive and immersive VR systems and enhancing human-computer interaction by exploring how sensory modalities and perspectives influence user engagement and interaction [18]. It will also shed light on the transferability of VR-acquired skills to real-world tasks, and this in turn may lead to improved training efficacy in critical fields such as surgery. By promoting accessible and inclusive iVLEs, this research extends its impact to communities and individuals with limited access to haptic devices, broadening its significance in advancing both theoretical knowledge and practical applications.

4 CONCLUSION

This paper describes a study that will be conducted to examine the effect of self-location, a sub-component of embodiment, by investigating the impact of first-person versus third-person perspectives on pseudo-haptics and performance during a virtual drilling task in VR. We hypothesize that the first-person perspective will result in increased drilling accuracy and enhanced perception of haptic feedback compared to the third-person perspective. Additionally, we anticipate reduced task completion times and fewer attempts in the first-person perspective. Embodiment is a critical factor in the effectiveness of psychomotor-based iVLEs, as acquiring new skills inherently requires physical presence to initiate, control, and refine movements. The integration of embodiment with pseudo-haptics is especially vital in contexts such as medical simulation and psychomotor skill education, where access to advanced haptic devices may be limited. This study aims to address gaps in understanding the role of embodiment in effective simulation-based training. We propose that pseudo-haptics can serve as an intermediary solution, bridging the gap between iVLEs designed for cognitive skill development and those offering realistic haptic feedback via high-fidelity, expensive devices.

ACKNOWLEDGMENTS

The financial support of the *Natural Sciences and Engineering Research Council of Canada* (NSERC) in the form of a *Discovery Grant* to B. K., *Ontario Tech University* in the form of a *Research Excellence Chair* to B. K., and the *Government of Ontario* in the form of a *Ontario Graduate Scholarship* to S. A., is gratefully acknowledged.

REFERENCES

[1] A. Adilkanov, M. Rubagotti, and Z. Kappasov. Haptic devices: Wearability-based taxonomy and literature review. *IEEE Access*,

10:91923–91947, Aug. 2022. doi: 10.1109/ACCESS.2022.3202986 1

[2] K. E. Adolph and J. E. Hoch. Motor development: embodied, embedded, enculturated, and enabling. *Annual Review of Psychology*, 70(1):141–164, Jan. 2019. doi: 10.1146/annurev-psych-010418-102836 1

[3] M. Berkman and G. Çatak. I-group presence questionnaire: Psychometrically revised english version. *Mugla Journal of Science and Technology*, 7:1–10, 2021. doi: 10.22531/muglajsci.882271 2

[4] I. d. V. Bosman, R. Tordiffe, T. J. Huang, O. O. Buruk, K. Jørgensen, and J. Hamari. Soundtrek: A virtual reality audio perspective-switching game. In *Companion Proceedings of the 2024 Annual Symposium on Computer-Human Interaction in Play*, CHI PLAY Companion '24, p. 352–358. Association for Computing Machinery, New York, NY, USA, 2024. doi: 10.1145/3665463.3678824 2

[5] Y. F. Cheng, L. M. Heller, S. Cho, and D. Lindlbauer. First or third-person hearing? a controlled evaluation of auditory perspective on embodiment and sound localization performance. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 81–90, 2024. doi: 10.1109/ISMAR62088.2024.00022 2, 3

[6] K. Collins and B. Kapralos. Pseudo-haptics: leveraging cross-modal perception in virtual environments. *The Senses and Society*, 14(3):313–329, Sept. 2019. doi: 10.1080/17458927.2019.1619318 1

[7] A. B. Cooper and E. J. Tisdell. Embodied aspects of learning to be a surgeon. *Medical Teacher*, 42(5):515–522, May 2020. doi: 10.1080/0142159X.2019.1708289 1

[8] S. G. Hart and L. E. Staveland. Development of nasa-tlx (task load index): Results of empirical and theoretical research. In P. A. Hancock and N. Meshkati, eds., *Human Mental Workload*, vol. 52 of *Advances in Psychology*, pp. 139–183. North-Holland, 1988. doi: 10.1016/S0166-4115(08)62386-9 2

[9] K. Kilteni, R. Groten, and M. Slater. The sense of embodiment in virtual reality. *Presence: Teleoperators and Virtual Environments*, 21(4):373–387, Nov. 2012. doi: 10.1162/PRES.a.00124 1, 2, 3

[10] D. Kirkpatrick and J. Kirkpatrick. *Transferring Learning to Behavior: Using the Four Levels to Improve Performance*. Berrett-Koehler Publishers, 2005. 2

[11] M. Liu, S. Fang, H. Dong, and C. Xu. Review of digital twin about concepts, technologies, and industrial applications. *Journal of Manufacturing Systems*, 58:346–361, 2021. Digital Twin towards Smart Manufacturing and Industry 4.0. doi: 10.1016/j.jmsy.2020.06.017 2

[12] A. Maselli and M. Slater. Sliding perspectives: dissociating ownership from self-location during full body illusions in virtual reality. *Frontiers in Human Neuroscience*, 8, Sept. 2014. doi: 10.3389/fnhum.2014.00693 3

[13] G. Ning, B. Grant, B. Kapralos, A. Quevedo, K. Collins, K. Kanev, and A. Dubrowski. Understanding virtual drilling perception using sound, and kinesthetic cues obtained with a mouse and keyboard. *Journal on Multimodal User Interfaces*, 17(3):165–165, Mar. 2023. doi: 10.1007/s12193-023-00412-x 1

[14] E. H. Peisachovich, A. Dubrowski, C. Da Silva, B. Kapralos, J. E. Klein, and Z. Rahmanov. Using simulation-based methods to support demonstration of competencies required by micro-credential courses. *Cureus*, 13(8):e16908, Aug. 2021. doi: 10.7759/cureus.16908 1

[15] F. G. Praticò, D. Calandra, M. Piviotti, and F. Lamberti. Assessing the user experience of consumer haptic devices for simulation-based virtual reality. In *2021 IEEE 11th International Conference on Consumer Electronics (ICCE-Berlin)*, pp. 1–6, 2021. doi: 10.1109/ICCE-Berlin53567.2021.9719998 1

[16] J. Sauro. *A Practical Guide to the System Usability Scale: Background, Benchmarks & Best Practices*. Measuring Usability LLC, 2011. 2

[17] F. J. Seagull and D. M. Rooney. Filling a void: Developing a standard subjective assessment tool for surgical simulation through focused review of current practices. *Surgery*, 156(3):718–722, 2014. doi: 10.1016/j.surg.2014.04.048 2

[18] A. M. Sofia Seinfeld, Tiare Feuchtnner and J. Müller. User representations in human-computer interaction. *Human-Computer Interaction*, 36(5-6):400–438, 2021. doi: 10.1080/07370024.2020.1724790 2, 3

[19] A. Toet, I. A. Kuling, B. N. Krom, and J. B. F. Van Erp. Toward

- enhanced teleoperation through embodiment. *Frontiers in Robotics and AI*, 7:14, Feb. 2020. doi: 10.3389/frobt.2020.00014 [1](#), [2](#)
- [20] Y. Ujitoko and Y. Ban. Survey of pseudo-haptics: Haptic feedback design and application proposals. *IEEE Transactions on Haptics*, 14(4):699–711, 2021. doi: 10.1109/TOH.2021.3077619 [1](#)